Rainfall Estimation

Contents

1 Introduction

Inverse-distance weighting (IDW) and related interpolation rules are widely used as simple baselines for rainfall-field estimation from microwave-link data. Their appeal is practical: they are fast, easy to implement, and often produce visually plausible maps. However, these methods do not explicitly search for a rainfall field whose forward-model-implied link attenuations match the observed link attenuations. This thesis investigates whether a nonlinear optimization model can improve on IDW and on link-aware ILDW when the goal is not only to predict rainfall, but also to remain faithful to the measurements that generated the estimate.

Contribution. This thesis makes one main contribution: it shows that the nonlinear optimization model outperforms IDW and ILDW in both fitting observed link attenuations and reducing rainfall estimation error, as presented through attenuation-error summaries and distance-binned rainfall-error summaries.

[Note: Consider restructuring this introduction following this structure: (i) goal of the thesis, (ii) contribution(s), and (iii) thesis structure, as in the style used in other master's theses.]

2 Background

[TODO: Write the Background section.]

3 Methodology

3.1 Data and Benchmark Construction

Our benchmark is built patch by patch. A *patch* is a rectangular crop of an hourly rainfall accumulation map from the OPERA European weather-radar product, selected to capture a detected rain event, with the patch width and height constrained to lie between 50 km and 95 km. For each patch in the benchmark list, we first extract the corresponding rainfall crop from the OPERA map. We then refine that crop from the native 2 km grid to a finer 125 m grid by splitting each coarse OPERA pixel into smaller subpixels and assigning each subpixel the parent pixel's rainfall value. We then smooth this refined field. This smoothed high-resolution rainfall field serves as ground truth for both simulating link attenuations and evaluating the solvers. The microwave-link data set, obtained from the 4TU-NL link data, is then overlaid on each patch, and only links whose endpoints fall inside the patch are retained. Each retained link preserves its original metadata (endpoints, frequency, and polarization) from the data set and is included in the benchmark only if both endpoints lie within the selected patch. Because the patch detector is designed to capture rainy events rather than dry background, the benchmark is rain-dominated: across the 100 patches, the mean non-rainy proportion is about 10.50 %, where non-rainy pixels are those with very little or no rainfall. Because the patch detector is designed to capture rainy events rather than dry background, the benchmark is rain-dominated: across the 100 patches, the mean non-rainy proportion is about 10.50 %, where non-rainy pixels are those with very little or no rainfall.

3.1.1 Geographic Distribution of the Benchmark Patches

Figure 1 maps the geographic distribution of the 100 benchmark patches. Each red rectangle marks the footprint of one selected patch on the Europe basemap. The figure makes clear that the benchmark is not concentrated in a single country or climatological regime: the selected rainy-event patches are spread across northern, western, central, southern, and southeastern Europe.

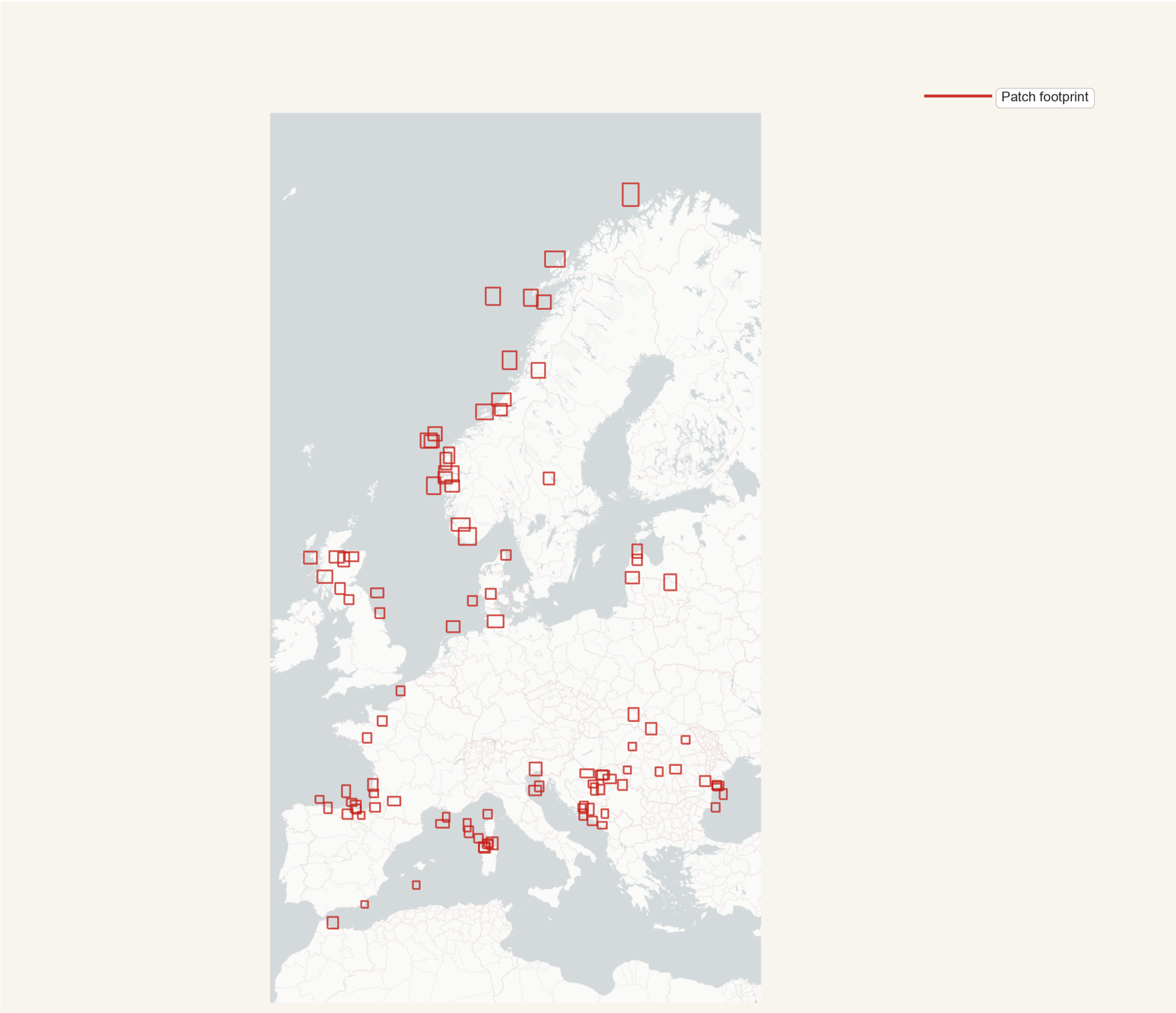

Figure 1: Geographic distribution of the 100 benchmark patches. Each patch is shown as a red outline rectangle over a Europe basemap. The figure visualizes the spatial spread of the selected rainy-event patches across the OPERA domain and provides geographic context for the benchmark used throughout the thesis.

3.1.2 Patch-Geometry Overview

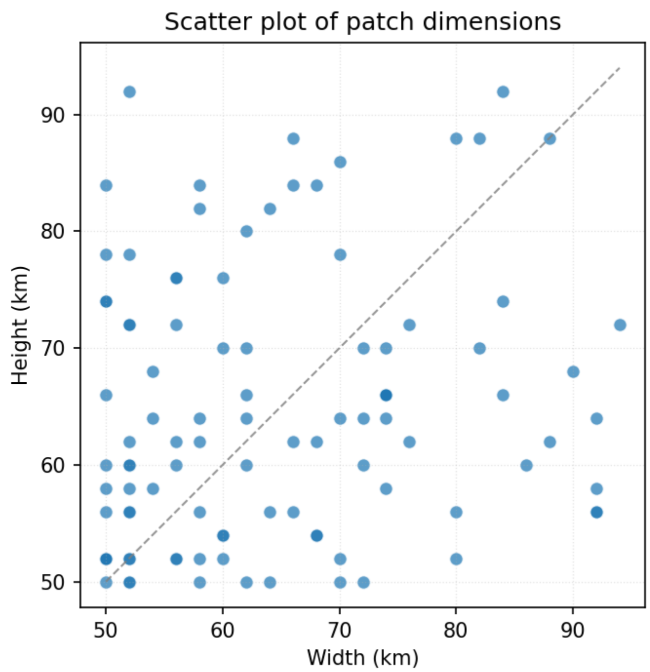

Figure 2: Patch width versus patch height for the 100 benchmark patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125 m).

Patch size (km)	# links	1-multiplicity	2-multiplicity
50×50	939	99	420
62×60	1254	128	563
72×70	1507	157	675

Table 1: Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

3.2 Forward Model

In the benchmark, link-level observations (attenuation and link metadata) are generated from the known ground-truth rainfall field via a forward model. Starting from the ground-truth rainfall map, each link is represented as a line segment between its endpoints, and attenuation is computed using ITU-R P.838-3, the standard relation for rain-induced specific attenuation in microwave links. In this model, the specific attenuation (dB/km) is

$$\gamma_R = \kappa(f, \mathrm{pol})\, R^{\alpha(f,\mathrm{pol})},$$

where R is rain rate (mm/h), and (κ, α) depend on link frequency and polarization. The total link attenuation is then the line integral of specific attenuation along the propagation path,

$$A_\ell \triangleq \int_\ell \gamma_R\big(R(s)\big)\,\mathrm{d}s.$$

In the benchmark, and likewise for the methods we compare against, this integral is evaluated in discretized form on the refined 125 m grid: each link is traced through crossed grid cells, and attenuation is accumulated as a length-weighted sum of cellwise specific attenuations,

$$A_\ell \approx \sum_{c \in \mathcal{C}(\ell)} \kappa_\ell \, R_c^{\alpha_\ell} \, \Delta s_{\ell,c},$$

where $\mathcal{C}(\ell)$ is the set of crossed cells and $\Delta s_{\ell,c}$ is the path length of link ℓ inside cell c (in km). These computed attenuation values, together with link metadata, form the link-level observations used for inverse reconstruction of the rainfall field.

3.3 Inverse Solvers

We compare three solvers: IDW, ILDW, and a nonlinear optimization model. Using the same link-level observations and forward-model conventions described above, we now describe each solver and its implementation details.

3.3.1 IDW

Inverse-distance weighting (IDW) is one of the standard simple interpolation strategies used in commercial-microwave-link rainfall reconstruction (Chwala and Kunstmann, 2019). The input consists of a set of links, where each link carries total attenuation, frequency, and polarization.

IDW first converts each observed link attenuation A_ℓ into a single link-level rain-rate estimate defined as:

$$R_\ell \triangleq \left(\frac{A_\ell/|\ell|}{\kappa_\ell}\right)^{1/\alpha_\ell}.$$

where A_ℓ denotes the attenuation of link ℓ, $|\ell|$ its length, and $(\kappa_\ell, \alpha_\ell)$ the ITU-R P.838-3 coefficients corresponding to its frequency and polarization.

Each per-link rain-rate estimate is then treated as a virtual point observation at the link midpoint. For each pixel p with center c_p, IDW computes a weighted average of nearby R_ℓ values, using weights proportional to $\|m_\ell - c_p\|_2^{-2}$ within a prescribed search radius $r_{\max}$ (set to 15km in our study). If one or more link midpoints lie inside pixel p, $\hat{R}_p$ (the estimated rainfall in pixel p), is set to the mean of their rainfall estimates. This defines the discretized IDW baseline used throughout this thesis.

Algorithm 1: Discretized IDW. Each link is replaced with a virtual gauge located at the midpoint of the link. The rain rate at the gauge is set to the uniform rain rate that reproduces the input link attenuation under ITU-R P.838-3. IDW is then applied with respect to these virtual gauges. We set the exponent of the decay to 2.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$

Result: rainfall field $\hat{R}$

foreach *link* $\ell \in \mathcal{L}$ **do** `// compute rain rate per link`

  $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$

  $\ell.\text{rainrate} \leftarrow \left(\dfrac{\ell.\text{attenuation}/\ell.\text{length}}{\ell.\kappa} \right)^{1/\ell.\alpha}$

foreach *pixel* $p \in P$ **do** `// compute rain` $\hat{R}_p$

  $M_p \leftarrow \{\ell \in \mathcal{L} : \ell.\text{midpoint} \in p\}$

  if $M_p \neq \emptyset$ **then**

    $\hat{R}_p \leftarrow \dfrac{1}{|M_p|} \sum_{\ell \in M_p} \ell.\text{rainrate}$ `// pixel contains a link endpoint`

  else

    $N_p \leftarrow \{\ell \in \mathcal{L} : \|\ell.\text{midpoint} - p.\text{center}\|_2 \leq r_{\max}\}$

    if $N_p = \emptyset$ **then**

      $\hat{R}_p \leftarrow 0$ `// pixel far from links`

    else

      $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \dfrac{\|\ell.\text{midpoint} - p.\text{center}\|_2^{-2}}{\sum_{\ell' \in N_p} \|\ell'.\text{midpoint} - p.\text{center}\|_2^{-2}} \cdot \ell.\text{rainrate}$

3.3.2 ILDW

ILDW is the link-aware variant introduced in this project as a stronger interpolation baseline. It starts from the same per-link rain-rate values obtained by inverting the discretized attenuation forward model for each link. The difference compared to IDW is the following: instead of treating each link as if all of its information were concentrated at the midpoint, ILDW uses the full link segment when deciding which links should influence a pixel and how strongly they should do so.

More precisely, for a pixel center c_p, ILDW replaces the midpoint distance $\|m_\ell - c_p\|_2$ by the point-to-segment distance

$$\text{dist}(c_p, \ell),$$

where $\ell = [p_{\ell,1}, p_{\ell,2}]$ denotes the full link segment with endpoint coordinates $p_{\ell,1}$ and $p_{\ell,2}$.

If one or more links cross the pixel, the estimate is taken to be the mean of the corresponding link rain-rate values. Otherwise, the estimate is a weighted average over nearby links within radius $r_{\max}$ (set to 15km), with weights proportional to the inverse squared segment distance. We considered this baseline important because it preserves the same simple interpolation philosophy as IDW while respecting more of the measurement geometry of the links themselves.

Algorithm 2: Discretized ILDW. Rainfall is averaged by inverse squared distance to a link. Lines that are different from their corresponding lines in IDW are colored red.

Data: link set $\mathcal{L}$, a patch P, and a maximum influence distance $r_{\max}$
Result: rainfall field $\hat{R}$

foreach *link* $\ell \in \mathcal{L}$ **do** `// compute rain rate per link`
  $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$
  $\ell.\text{rainrate} \leftarrow \left(\dfrac{\ell.\text{attenuation}/\ell.\text{length}}{\ell.\kappa}\right)^{1/\ell.\alpha}$
foreach *pixel* $p \in P$ **do** `// compute rain` $\hat{R}_p$
  $X_p \leftarrow \{\ell \in \mathcal{L} : \ell \cap p \neq \emptyset\}$
  if $X_p \neq \emptyset$ **then**
    $\hat{R}_p \leftarrow \dfrac{1}{|X_p|} \sum_{\ell \in X_p} \ell.\text{rainrate}$ `// at least one link traverses the pixel`
  else
    $N_p \leftarrow \{\ell \in \mathcal{L} : \text{dist}(p, \ell) \leq r_{\max}\}$
    if $N_p = \emptyset$ **then**
      $\hat{R}_p \leftarrow 0$ `// pixel far from links`
    else
      $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \dfrac{\text{dist}^{-2}(p.\text{center}, \ell)}{\sum_{\ell' \in N_p} \text{dist}^{-2}(p.\text{center}, \ell)} \cdot \ell.\text{rainrate}$

3.3.3 Nonlinear Optimization Method

The nonlinear optimizer is the method we introduced to address some of the limitations of IDW and ILDW. Both interpolation-based baselines first reduce each link to a link-level rain estimate and then spread that information spatially by geometric weighting.

The optimizer used in this study is L-BFGS-B, a limited-memory quasi-Newton method for smooth constrained optimization (**??**). In the present setting, the optimization variable is the rainfall value at every pixel of the patch, so the problem dimension is the number of pixels in the patch. Rather than storing a full Hessian matrix, L-BFGS-B builds a low-memory approximation to local curvature information from the most recent iterates and gradients, and uses that approximation to propose a descent direction. A line search is then carried out along that direction, and the process is repeated until the stopping criteria on iteration count, function reduction, or projected gradient norm are met. The suffix "-B" indicates that the method handles box constraints. Here those constraints enforce nonnegativity of the rainfall field. The optimizer is initialized from the ILDW rainfall estimate because the nonlinear objective explicitly seeks to fit the observed link attenuations, and ILDW is constructed from those same attenuation measurements and the full link geometry. For isolated links, ILDW selects the per-link rain rate so that reapplying the forward model to those links alone reproduces the observed attenuation. This attenuation-level consistency makes ILDW a convenient warm start for the nonlinear optimization.

Optimization Objective For each patch P, we identify P with its set of pixels, and denote by $\mathcal{L}(P)$ the set of links contained in P. Let $r \in \mathbb{R}_{\geq 0}^{|P|}$ denote the unknown rainfall field written as a vector over pixels. We solve the constrained optimization problem

$$\min_{r \in \mathbb{R}_{\geq 0}^{|P|}} J(r),$$

where the objective is a weighted sum of four terms:

$$J(r) \triangleq \alpha_{\text{atten}} J_{\text{atten}}(r) \;+\; \alpha_{1d} J_{1d}(r) \;+\; \alpha_{2d} J_{2d}(r) \;+\; \alpha_{\text{total}} J_{\text{total}}(r).$$

Here J_{atten} penalizes disagreement between observed and forward-model-predicted link attenuations, while J_{1d}, J_{2d}, and J_{total} are regularization terms that encourage the estimated field to remain spatially and physically well behaved. In the normalized solver studied here, we define the weights instance by instance using the corresponding objective values at the ILDW solution:

$$\alpha_{\text{atten}} \triangleq \frac{1}{J_{\text{atten}}(\text{ILDW})}, \qquad \alpha_{1d} \triangleq \frac{1}{J_{1d}(\text{ILDW})}, \qquad \alpha_{2d} \triangleq \frac{1}{2} \cdot \frac{1}{J_{2d}(\text{ILDW})}, \qquad \alpha_{\text{total}} \triangleq \frac{1}{J_{\text{total}}(\text{ILDW})}.$$

More concretely, J_{atten} is the attenuation-misfit term:

$$J_{\text{atten}}(r) \triangleq \frac{1}{|\mathcal{L}(P)|} \sum_{\ell \in \mathcal{L}(P)} \frac{\left(\widehat{A}_\ell(r) - A_\ell^{\text{obs}}\right)^2}{|\ell|},$$

where $\widehat{A}_\ell(r)$ is the attenuation predicted by the forward model from the rainfall field r, A_ℓ^{obs} is the observed attenuation of link ℓ, and $|\ell|$ is the link length in km. Thus J_{atten} measures how far the forward-model-implied attenuations are from the observed attenuations, with a normalization by link length and averaging over the links in the patch.

The term J_{1d} is a first-difference penalty:

$$J_{1d}(r) \triangleq \frac{1}{|P|} \sum_{(u,v) \in \mathcal{E}(P)} \left(r_u - r_v\right)^2,$$

where $\mathcal{E}(P)$ is the set of horizontally and vertically adjacent pixel pairs in patch P. This term sums squared differences between neighboring pixel values and therefore discourages abrupt pixel-to-pixel jumps in the estimated rainfall field.

The term J_{2d} is a second-difference penalty:

$$J_{2d}(r) \triangleq \frac{1}{|P|} \sum_{(p,q,t) \in \mathcal{T}(P)} \left((r_q - r_p) - (r_t - r_q)\right)^2,$$

where $\mathcal{T}(P)$ is the set of horizontal and vertical triples of collinear pixels in patch P. It penalizes changes in the first difference, so it discourages unnecessary curvature or oscillation along rows and columns.

Finally, J_{total} is an ℓ_2 shrinkage term on the rainfall field itself:

$$J_{\text{total}}(r) \triangleq \frac{1}{|P|} \sum_{p \in P} r_p^2.$$

This term biases the solution toward smaller overall rainfall values, except where larger values are needed to match the observed link attenuations.

These four terms are added together only after weighting, so the optimizer balances attenuation fit against smoothness, curvature control, and shrinkage.

4 Evaluation

4.1 Evaluation Protocol

We evaluate each solver in four ways: overall agreement with the ground-truth rainfall map, how errors change as distance from links increases, wet/dry classification behavior, and agreement with observed link attenuations under the forward model. We first report patch-level summaries across all 100 benchmark patches, and then provide qualitative case studies in the appendix.

Key findings at a glance. Across the benchmark, the nonlinear optimizer gives the highest overall agreement with the ground-truth rainfall map and the best attenuation consistency. Distance-conditioned analyses also show that all methods degrade as link support becomes weaker, but the optimizer and ILDW remain more robust than IDW in sparsely constrained regions.

4.2 Global Patch-Level Metrics

We now turn to the empirical comparison of the three methods on the hundred-patch benchmark. We begin with whole-patch summaries, then examine how errors vary with link proximity, and finally report attenuation-consistency diagnostics.

Before turning to distance-binned diagnostics, it is useful to summarize each predicted rainfall map at the level of an entire patch. For a patch P and solver s, we therefore compute three scalar metrics over all pixels in P: the root mean squared error

$$\mathrm{RMSE}_{P,s} \triangleq \sqrt{\frac{1}{|P|}\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - R_P(p)\big)^2},$$

the signed bias

$$\mathrm{Bias}_{P,s} \triangleq \frac{1}{|P|}\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - R_P(p)\big),$$

and the Pearson correlation coefficient

$$r_{P,s} \triangleq \frac{\mathrm{cov}_P\big(\widehat{R}_{P,s}, R_P\big)}{\sigma_P\big(\widehat{R}_{P,s}\big)\,\sigma_P\big(R_P\big)},$$

where

$$\mathrm{cov}_P\big(\widehat{R}_{P,s}, R_P\big) \triangleq \frac{1}{|P|}\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s}\big)\big(R_P(p) - \overline{R}_P\big),$$

$$\sigma_P\big(\widehat{R}_{P,s}\big) \triangleq \sqrt{\frac{1}{|P|}\sum_{p\in P}\big(\widehat{R}_{P,s}(p) - \overline{\widehat{R}}_{P,s}\big)^2}, \quad \sigma_P\big(R_P\big) \triangleq \sqrt{\frac{1}{|P|}\sum_{p\in P}\big(R_P(p) - \overline{R}_P\big)^2},$$

with

$$\overline{\widehat{R}}_{P,s} \triangleq \frac{1}{|P|}\sum_{p\in P}\widehat{R}_{P,s}(p), \quad \overline{R}_P \triangleq \frac{1}{|P|}\sum_{p\in P}R_P(p).$$

RMSE measures overall magnitude error. Bias measures average signed error (net over- or underestimation). Correlation measures how well the spatial structure of the field is reproduced independently of scale and offset. The Pearson correlation coefficient satisfies $r_{P,s} \in [-1, 1]$. A value of 1 indicates perfect positive linear agreement between predicted and true rainfall maps. A

Solver	RMSE	Bias	Correlation
IDW	1.307 ± 1.234	0.034 ± 0.296	0.866 ± 0.059
ILDW	1.239 ± 1.182	0.047 ± 0.261	0.878 ± 0.055
Nonlinear optimizer	1.196 ± 1.179	-0.186 ± 0.335	0.901 ± 0.055

Table 2: Patch-level summary of three global map metrics across the 100 evaluated patches. Each entry reports mean $\pm$ standard deviation across patches. Here RMSE and Bias are computed in the same rainfall units as R_P and $\widehat{R}_{P,s}$, while Correlation is the Pearson correlation coefficient between the predicted and ground-truth rainfall fields within each patch. The corresponding RMSE ranges are $[0.247, 6.932]$ for IDW, $[0.229, 7.062]$ for ILDW, and $[0.176, 6.940]$ for the nonlinear optimizer.

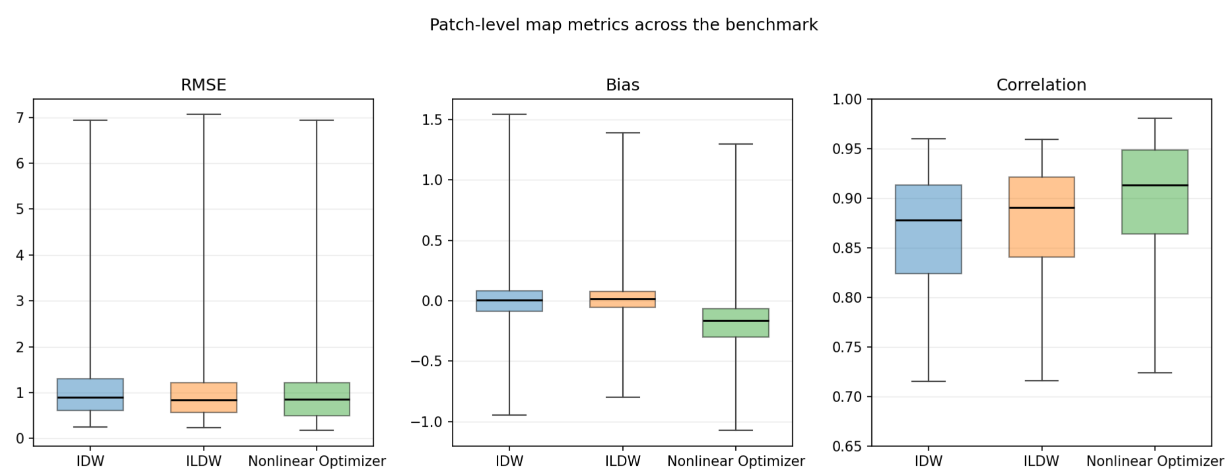

Figure 3: Patch-level distributions of RMSE, Bias, and Pearson correlation coefficient across the 100 evaluated patches. Each panel shows one metric, with one boxplot per solver. The RMSE panel summarizes overall magnitude error, the Bias panel summarizes average signed error (net over- or underestimation), and the Correlation panel summarizes how well each method reproduces the spatial rainfall pattern.

value of 0 indicates no linear relationship. A value of -1 indicates perfect negative linear agreement (an exact inverse pattern).

Overall, these global metrics indicate that the nonlinear optimizer best preserves patch-scale spatial structure while reducing aggregate map error.

4.2.1 Link-Consistency Objective (J_{atten})

Solver	Mean J_{atten} (dB^2/km)	SD
IDW	0.01522	0.03808
ILDW	0.00289	0.00757
Nonlinear Optimizer	0.00020	0.00103

Table 3: Patch-level summary of J_{atten} across the 100 evaluated patches. For a given patch, J_{atten} is the attenuation-fit term $J_{\text{atten}} = \frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{(\hat{A}_\ell - A_\ell^{\text{obs}})^2}{|\ell|}$, where $\hat{A}_\ell$ is the attenuation implied by the estimated rainfall field on link ℓ, A_ℓ^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. The table reports the mean and standard deviation of this per-patch quantity for each solver. Lower values indicate better agreement with the observed link attenuations.

4.2.2 Absolute Attenuation Error per Kilometer

Solver	Mean (dB/km)	SD
IDW	0.05034	0.03413
ILDW	0.01356	0.00948
Nonlinear Optimizer	0.00230	0.00437

Table 4: Patch-level summary of the mean absolute attenuation error per km. For a given patch, we compute $\frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{|\hat{A}_\ell - A_\ell^{\text{obs}}|}{|\ell|}$, where $\hat{A}_\ell$ is the attenuation induced by the estimated rainfall field on link ℓ, A_ℓ^{obs} is the observed attenuation, and $|\ell|$ is the link length in kilometers. Thus, the absolute attenuation error of every link is normalized by the link's length. The table reports the mean and standard deviation of this quantity across the 100 evaluated patches for each solver. Lower values indicate better agreement with the observed link attenuations.

Together, the attenuation-based diagnostics show that the nonlinear optimizer most consistently matches the measured link attenuations under the shared forward model.

4.3 Distance-Conditioned Error Analysis

We next ask how prediction quality varies with link proximity and how well the predicted rainfall fields explain the observed link measurements. To answer these questions, we first group pixels into distance bins according to their distance to the third-closest link and summarize patch-level error statistics within each bin. We first examine rainy pixels through relative absolute error, and then examine non-rainy pixels through absolute error. In this subsection we focus exclusively on distance-conditioned pixelwise diagnostics.

4.3.1 Rainy Pixel Relative Absolute Error vs. Link Distance

For a patch P, let $R_P : P \to \mathbb{R}_{\geq 0}$ be the ground-truth rainfall field, and let $\widehat{R}_{P,s} : P \to \mathbb{R}_{\geq 0}$ be the rainfall field predicted by solver s. We define a pixel p to be rainy if $R_P(p) \geq 0.6$ mm/h, and we write

$$P_{\text{rain}} = \{p \in P : R_P(p) \geq 0.6\}.$$

For a rainy pixel p, the relative absolute error with respect to solver s is

$$\text{RAE}_{P,s}(p) \triangleq \frac{|R_P(p) - \widehat{R}_{P,s}(p)|}{R_P(p)}.$$

To study how prediction quality changes with link proximity, we group pixels by distance to the third-closest link. For a pixel $p \in P$, let $c(p) \in \mathbb{R}^2$ denote the center of pixel p. For a link $\ell \in \mathcal{L}(P)$, viewed geometrically as a line segment, let

$$\text{dist}(p, \ell) \triangleq \text{dist}\big(c(p), \ell\big)$$

denote the Euclidean distance from the pixel center to that link segment. If the links in patch P are written as $\ell_1, \ldots, \ell_{|\mathcal{L}(P)|}$, we define

$$d_3(p) \triangleq \text{the third-order statistic of } \big\{\text{dist}(p, \ell_j)\big\}_{j=1}^{|\mathcal{L}(P)|},$$

that is, the distance from the center of pixel p to its third-closest link segment.

For each index i, let b_{i-1} and b_i denote the lower and upper distance cutoffs of the ith bin, and write $B_i = (b_{i-1}, b_i]$. A pixel p belongs to bin B_i exactly when $d_3(p) \in B_i$.

For each patch P, solver s, and distance bin B_i, we then collect the rainy-pixel RAE values in that bin and summarize them by the patch-level median

$$m_{s,i}^{\text{rain}}(P) \triangleq \text{median}\big\{\text{RAE}_{P,s}(p) : p \in P_{\text{rain}},\ d_3(p) \in B_i\big\} .$$

In Figure 4, these patch-level medians are compared across the full benchmark collection $\mathcal{P}$ of 100 patches. There, the displayed distance bins are

$$[0, 125] \text{ m},\ (125, 375] \text{ m},\ (375, 750] \text{ m},\ (750, 1500] \text{ m},\ (1500, 3125] \text{ m},\ (3125, 6000] \text{ m},$$

with the sparsely populated tail beyond 6000 m merged into the last displayed bin. For fixed solver s and distance bin B_i, let

$$\mu_{s,i}^{\text{rain}} \triangleq \{m_{s,i}^{\text{rain}}(P)\}_{P \in \mathcal{P}}$$

denote the sequence of patch-level rainy-pixel median RAE values across patches. Each box-and-whisker then summarizes this sequence: the lower whisker tip is the minimum of $\mu_{s,i}^{\text{rain}}$, the lower edge of the box is its 25th percentile, the horizontal line segment with the filled circle at its center marks the median, the upper edge of the box is its 75th percentile, and the upper whisker tip is its maximum.

4.3.2 Non-Rainy Pixel Absolute Error vs. Link Distance

For the dry-pixel analysis, we focus on pixels whose ground-truth rainfall falls below the wet threshold. We therefore define the non-rainy pixel set by

$$P_{\text{nonrain}} = \{p \in P : R_P(p) < 0.6\}.$$

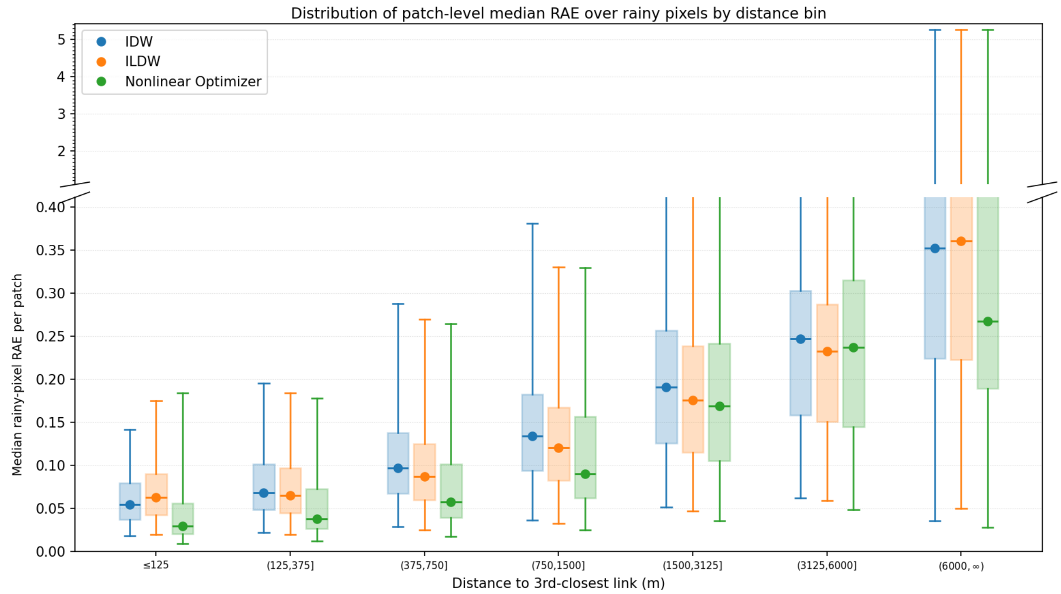

Figure 4: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a rainy pixel is a pixel whose ground-truth rainfall is at least the configured wet threshold (set to 0.6 mm/h), and the relative absolute error for a rainy pixel p by solver s is defined as $|R_P(p) - \widehat{R}_{P,s}(p)|/R_P(p)$. The corresponding rainy-pixel counts per bin are reported in Table 5.

Distance to third closest link (m)	Rainy mean count (% of total)	SD
$[0, 125]$	4267.0 (1.79%)	960.4
$(125, 375]$	16737.7 (7.01%)	3803.2
$(375, 750]$	26809.9 (11.22%)	6707.0
$(750, 1500]$	51988.5 (21.76%)	14 468.0
$(1500, 3125]$	88839.4 (37.18%)	27 531.1
$(3125, 6000]$	45929.4 (19.22%)	15 679.4
$(6000, 9000]$	3835.8 (1.61%)	2725.7
$(9000, \infty)$	514.4 (0.22%)	966.9
Total	238922.1 (100%)	

Table 5: Rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 4. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the rainy-pixel count in each bin across patches. The last bin, $(9000, \infty)$, is merged with the previous bin in Figure 4 because it is sparsely populated.

For these pixels, relative normalization by $R_P(p)$ is not appropriate because the denominator may be zero or very small. We therefore measure prediction quality by absolute error rather than relative absolute error:

$$\mathrm{AE}_{P,s}(p) = |R_P(p) - \widehat{R}_{P,s}(p)|.$$

Using the same definition of the third-closest-link distance $d_3(p)$ introduced above, and the same distance bins B_i as in the rainy-pixel analysis, we compute for each patch P, solver s, and distance bin B_i

$$m_{s,i}^{\text{nonrain}}(P) \triangleq \text{median}\big\{\mathrm{AE}_{P,s}(p) : p \in P_{\text{nonrain}},\ d_3(p) \in B_i\big\},$$

and, for fixed s and i, the corresponding box-and-whisker plot summarizes the sequence

$$\mu_{s,i}^{\text{nonrain}} \triangleq \{m_{s,i}^{\text{nonrain}}(P)\}_{P\in\mathcal{P}}$$

of patch-level non-rainy-pixel median AE values across patches. Thus the rainy and non-rainy figures use the same distance-bin construction, but they differ both in the pixel set being summarized and in the error metric being aggregated.

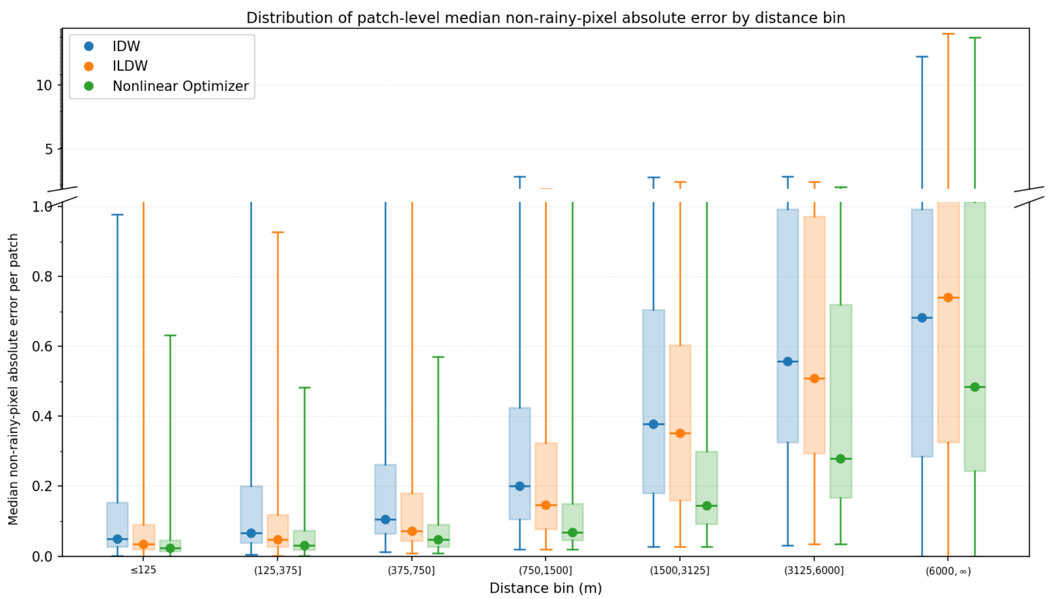

Figure 5: Non-rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median non-rainy-pixel absolute error value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a non-rainy pixel is a pixel whose ground-truth rainfall is below the configured wet threshold (set to 0.6 mm/h), and the absolute error for a non-rainy pixel p by solver s is defined as $|R_P(p) - \widehat{R}_{P,s}(p)|$. The corresponding non-rainy-pixel counts per bin are reported in Table 6.

Distance to third closest link (m)	Non-rainy mean count (% of total)	SD
$[0, 125]$	563.6 (1.84%)	777.6
$(125, 375]$	2261.1 (7.40%)	3093.5
$(375, 750]$	3389.1 (11.09%)	4709.5
$(750, 1500]$	6446.5 (21.09%)	9257.7
$(1500, 3125]$	10776.2 (35.25%)	15 553.7
$(3125, 6000]$	6197.2 (20.27%)	8990.0
$(6000, 9000]$	849.9 (2.78%)	1289.7
$(9000, \infty)$	85.5 (0.28%)	250.9
Total	30569.2 (100%)	

Table 6: Non-rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 5. For each patch, pixels are classified by the value of $d_3(p)$, and the table reports the mean and standard deviation of the non-rainy-pixel count in each bin across patches. The last bin, $(9000, \infty)$, is merged with the previous bin in Figure 5 because it is sparsely populated.

Across distance bins, performance differences widen as pixels become less constrained by nearby links, which motivates the geometry-focused context shown next.

4.3.3 Largest Patch: Link-Geometry Context

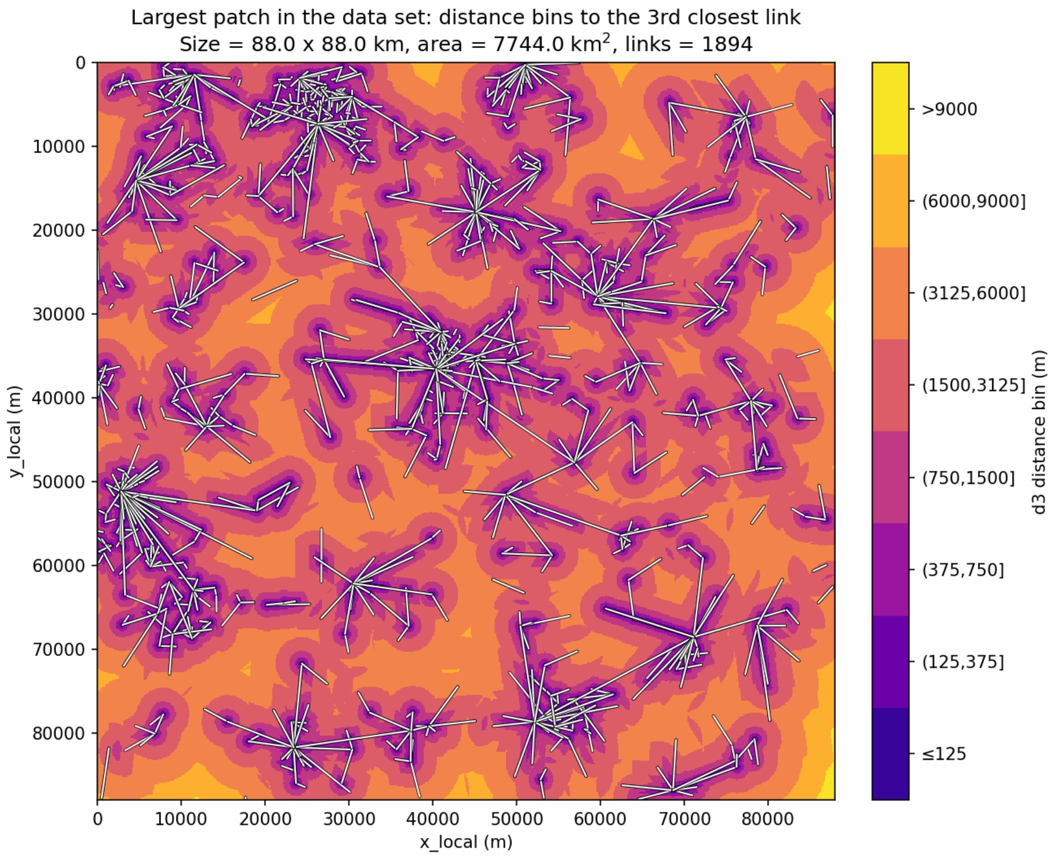

Figure 6: Distance-bin map for the largest evaluated patch in the data set. Each pixel is colored according to the bin induced by its distance to the third closest link, and the complete link geometry is overlaid in white.

The same largest patch is also useful for describing the structure of the underlying link network. In the summaries below, link length refers to the path length of a link, carrier frequency is the microwave-link frequency in GHz, and endpoint degree is the number of links incident to a given endpoint node. Here, an endpoint node is a location at which one or more links meet.

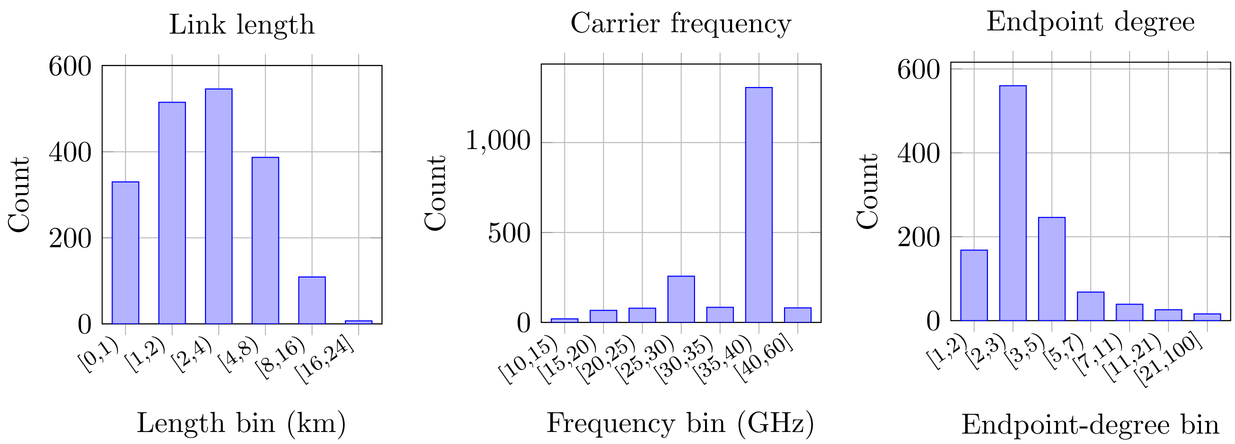

Figure 7: Histogram summaries for the largest patch. Left: link lengths. Middle: carrier frequencies. Right: endpoint-degree histogram. The x-axis gives endpoint degree and the y-axis gives the number of endpoint nodes in each degree bin.

Length bin (km)	Count	%
$[0, 1)$	330	17.4
$[1, 2)$	515	27.2
$[2, 4)$	546	28.8
$[4, 8)$	387	20.4
$[8, 16)$	109	5.8
$[16, 24]$	7	0.4

Frequency bin (GHz)	Count	%
$[10, 15)$	20	1.1
$[15, 20)$	67	3.5
$[20, 25)$	79	4.2
$[25, 30)$	257	13.6
$[30, 35)$	84	4.4
$[35, 40)$	1306	69.0
$[40, 60]$	81	4.3

Endpoint-degree bin	Count	%
$[1, 2)$	168	15.0
$[2, 3)$	560	49.9
$[3, 5)$	246	21.9
$[5, 7)$	68	6.1
$[7, 11)$	39	3.5
$[11, 21)$	26	2.3
$[21, 100]$	16	1.4

Table 7: Tabulated distributions for the largest-patch link network. The first panel counts links by path-length interval, the second counts links by carrier-frequency interval, and the third counts unique endpoint nodes by their graph degree.

4.3.4 Confusion Matrix

IDW

	Prediction	
GT	Wet	Dry
Wet	0.893 ± 0.013	0.002 ± 0.000
Dry	0.040 ± 0.004	0.065 ± 0.010

ILDW

	Prediction	
GT	Wet	Dry
Wet	0.892 ± 0.013	0.003 ± 0.000
Dry	0.037 ± 0.004	0.068 ± 0.010

Nonlinear Optimizer

	Prediction	
GT	Wet	Dry
Wet	0.890 ± 0.013	0.005 ± 0.001
Dry	0.030 ± 0.004	0.075 ± 0.010

Table 8: Patch-averaged wet/dry confusion matrices for IDW, ILDW, and the nonlinear optimizer at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels: top-left is TP, top-right is FN, bottom-left is FP, and bottom-right is TN. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM).

5 Related Work

[TODO: Write the Related Work section.]

6 Conclusion and Outlook

[TODO: Write the Conclusion and Outlook section.]

7 Appendix

7.1 Case Study Figures and Optimizer Traces

The five case studies below pair a geographic view of the selected patch with the corresponding rainy-pixel solver outputs. The timestamp in each caption identifies the radar scene, and patch 0 denotes the selected patch within that scene. This layout is intended to keep the geographic context and the solver behavior visually tied to one another.

These five cases were selected to span distinct rainfall morphologies and link-coverage configurations, so that qualitative comparisons can reveal where each method tends to oversmooth, displace, or better preserve localized rain structures.

The optimizer-trace plots included below show, for each selected case study, how the weighted objective and its weighted constituent terms decrease over the optimization iterations.

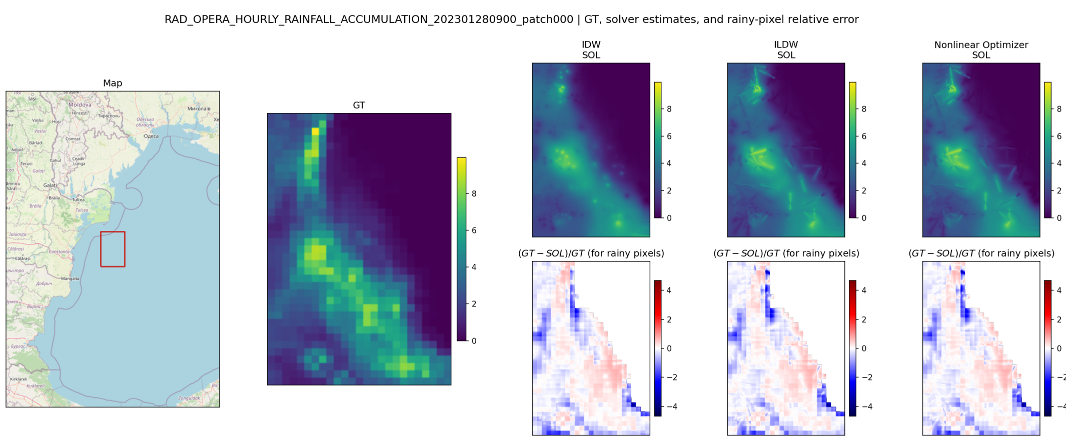

Figure 8: Case study for a rain event observed on January 28, 2023 at 09:00, for a patch located near the western Black Sea coast in southeastern Romania. The combined panel shows the geographic map of the selected patch, the ground-truth rainfall field R_P, and, for each solver, the corresponding estimate $\widehat{R}_{P,s}$ together with the rainy-pixel relative error $(R_P - \widehat{R}_{P,s})/R_P$.

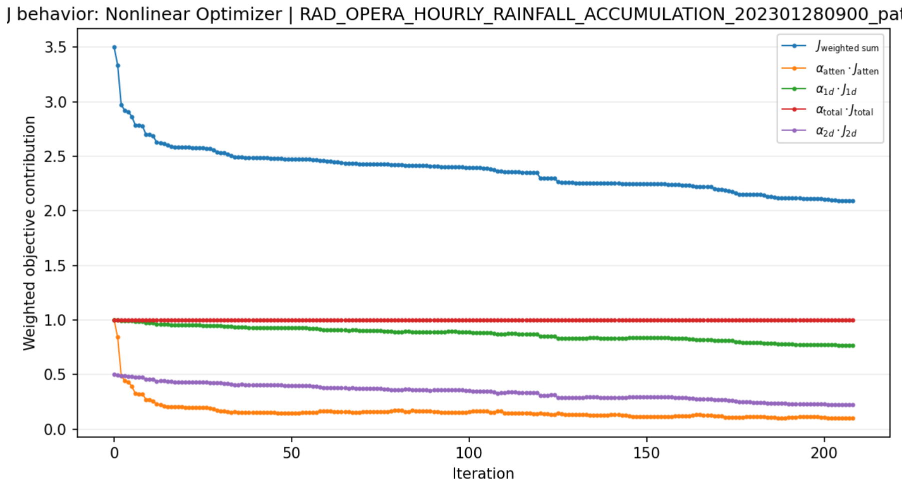

Figure 9: Optimizer trace for the January 28, 2023 case study at 09:00. The plot shows the reduction of the weighted objective $J_{\text{weighted sum}}$ and of the weighted constituent terms over L-BFGS-B iterations.

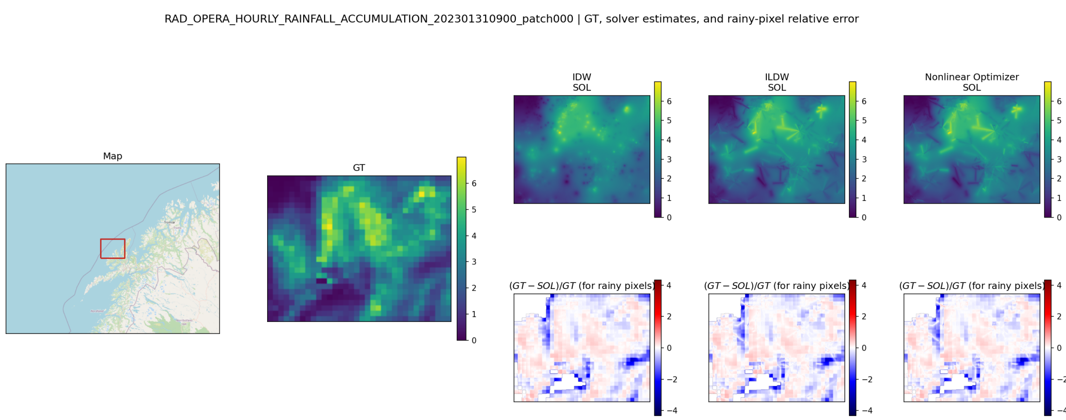

Figure 10: Case study for a rain event observed on January 31, 2023 at 09:00, for a patch located in northern Norway. The combined panel shows the geographic map of the selected patch, the ground-truth rainfall field R_P, and, for each solver, the corresponding estimate $\widehat{R}_{P,s}$ together with the rainy-pixel relative error $(R_P - \widehat{R}_{P,s})/R_P$.

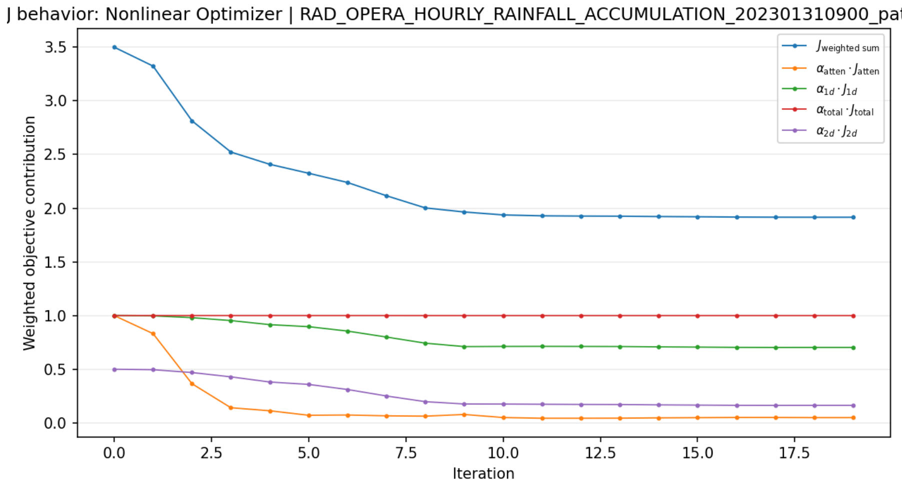

Figure 11: Optimizer trace for the January 31, 2023 case study at 09:00. The plot reports the iteration-by-iteration decrease of the weighted objective and its weighted constituent terms.

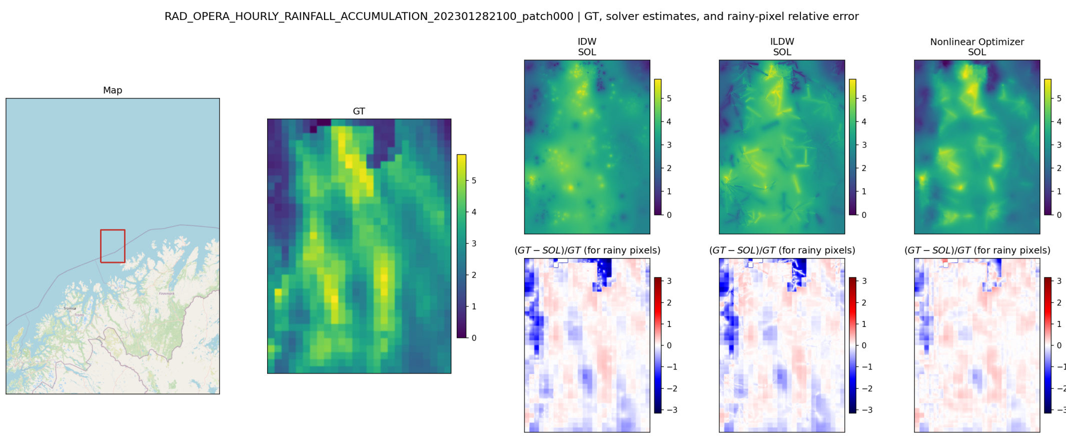

Figure 12: Case study for a rain event observed on January 28, 2023 at 21:00, for a patch located in far northern Norway near the Barents Sea. The combined panel shows the geographic map of the selected patch, the ground-truth rainfall field R_P, and, for each solver, the corresponding estimate $\widehat{R}_{P,s}$ together with the rainy-pixel relative error $(R_P - \widehat{R}_{P,s})/R_P$.

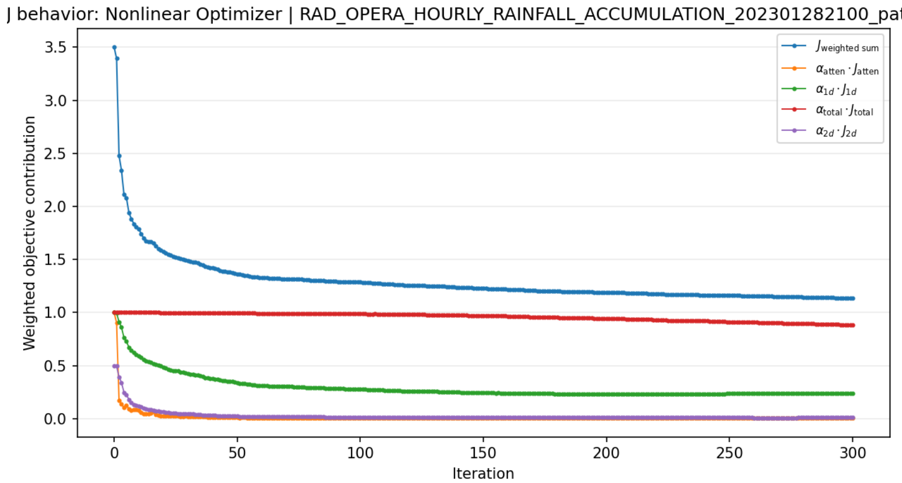

Figure 13: Optimizer trace for the January 28, 2023 case study at 21:00. The plot shows the reduction of the weighted objective $J_{\text{weighted sum}}$ and of the weighted constituent terms over L-BFGS-B iterations.

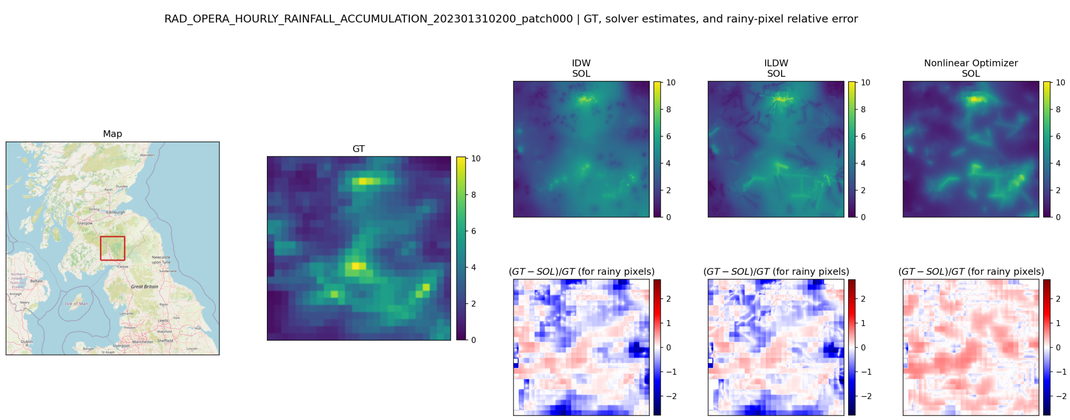

Figure 14: Case study for a rain event observed on January 31, 2023 at 02:00, for a patch located in southern Scotland near Edinburgh. The combined panel shows the geographic map of the selected patch, the ground-truth rainfall field R_P, and, for each solver, the corresponding estimate $\widehat{R}_{P,s}$ together with the rainy-pixel relative error $(R_P - \widehat{R}_{P,s})/R_P$.

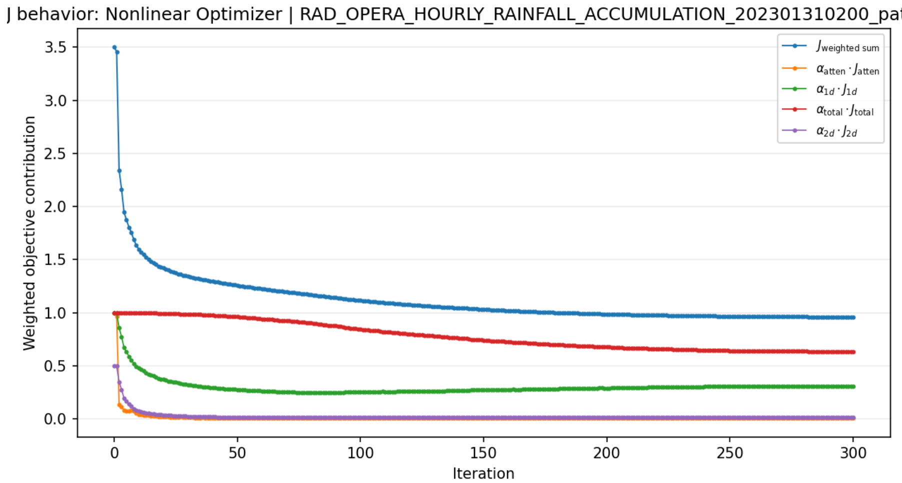

Figure 15: Optimizer trace for the January 31, 2023 case study at 02:00. The plot reports the iteration-by-iteration decrease of the weighted objective and its weighted constituent terms.

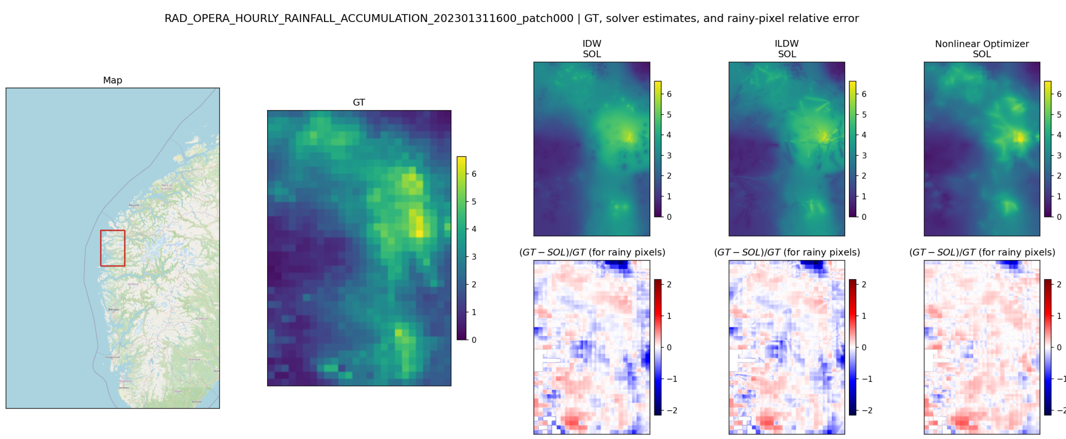

Figure 16: Case study for a rain event observed on January 31, 2023 at 16:00, for a patch located on the western coast of Norway. The combined panel shows the geographic map of the selected patch, the ground-truth rainfall field R_P, and, for each solver, the corresponding estimate $\widehat{R}_{P,s}$ together with the rainy-pixel relative error $(R_P - \widehat{R}_{P,s})/R_P$.

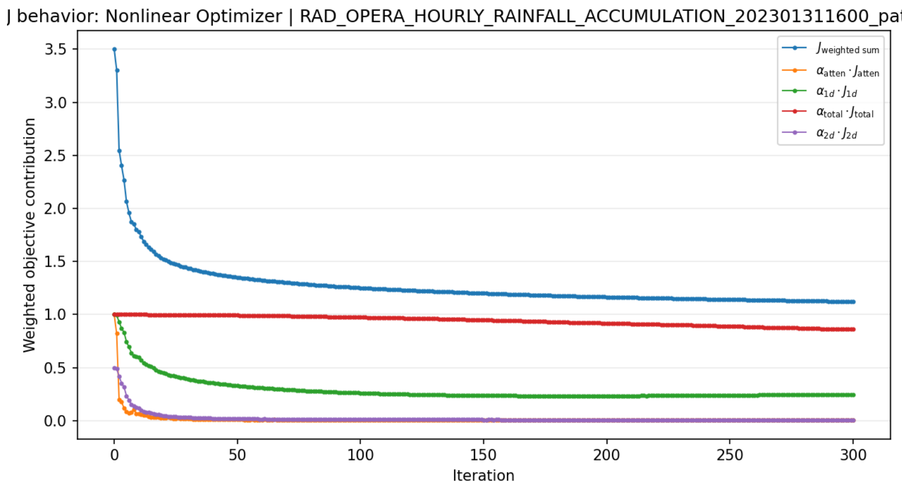

Figure 17: Optimizer trace for the January 31, 2023 case study at 16:00. The weighted objective and weighted component terms are shown over the L-BFGS-B iterations for the same patch displayed in the spatial case-study panels above.

References

Christian Chwala and Harald Kunstmann. Commercial microwave link networks for rainfall observation: assessment of the current status and future challenges. *WIREs Water*, 6(2):e1337, 2019. doi: 10.1002/wat2.1337.